The Journal of Open Source Software

part2pop: a modular framework for standardizing and analyzing aerosol particle populations

Laura Fierce [1] **and Payton Beeler** [1]

1 Pacific Northwest National Laboratory

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Open Journals

Reviewers:

- @openjournals

Submitted: 01 January 1970
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

Aerosol particles are chemically diverse, multicomponent systems that vary widely in size, composition, and mixing state. These particle-scale differences strongly influence cloud condensation nuclei activity, ice nucleation, and optical properties (Alpert & Knopf, 2016; Cubison et al., 2008; Fierce et al., 2013, 2016, 2020; Lata et al., 2021; Petters & Kreidenweis, 2007; Riemer et al., 2019).

Across models, observations, and parameterized descriptions, aerosol populations are represented in incompatible ways. Different representations preserve different aspects of aerosol size, composition, and mixing state, which limits direct comparison across representations (Fierce et al., 2024; Liu et al., 2016; Riemer et al., 2009; Stevens & Dastoor, 2019; Tilmes et al., 2023). As a result, similar analyses are frequently reimplemented for each representation, making comparison and reuse time-consuming.

`part2pop` is a lightweight Python library that standardizes heterogeneous aerosol populations into a common data model. The package supports population construction from model output, parameterizations, and observation-constrained reconstructions, and it provides reusable particle- and population-level diagnostics for aerosol composition, hygroscopicity, cloud activation, optical properties, and freezing. Its registry-based builder architecture makes the package extensible, allowing new species definitions, population types, particle morphologies, and diagnostic variables to be added without modifying the core representation.

Statement of need

Aerosol populations from particle-resolved models, reduced representations such as modal and sectional models, parameterized distributions, and observation-based reconstructions represent different information about particle size, composition, abundance, and mixing state (Bauer et al., 2008; DeCarlo et al., 2006; Jayne et al., 2000; Liu et al., 2016; Mann et al., 2012; Riemer et al., 2009; Stevens & Dastoor, 2019; Tilmes et al., 2023; Wang, Pikridas, Spielman, et al., 2017; Zaveri et al., 2008; Zelenyuk et al., 2015). Consequently, these populations cannot be directly compared without translation, and analyses such as composition metrics, CCN activity, optical properties, and freezing diagnostics are often reimplemented for each representation.

`part2pop` addresses this gap by providing a shared aerosol-population data model and analysis interface that makes such analyses reusable across heterogeneous sources. The package does not introduce a new aerosol simulation model. Instead, it standardizes populations derived from simulations, observations, and parameterizations so that they can be analyzed, compared, and reused consistently. Intended users include aerosol model developers, instrument developers and observationalists, process modelers requiring aerosol inputs, instructors and students, and researchers performing intercomparisons or sensitivity analyses. The need addressed is not a lack of aerosol microphysics models, but a lack of a common internal representation of

aerosol populations that makes existing tools reusable across models, parameterizations, and observations.

State of the field

Existing software in aerosol science typically focuses on either source-specific aerosol populations or targeted downstream calculations, rather than interoperability across representations. Aerosol simulation frameworks generate populations in source-specific representations, reflecting trade-offs between physical detail and computational efficiency (Liu et al., 2016; Mann et al., 2012; Riemer et al., 2009; Tilmes et al., 2023). Observational workflows constrain different subsets of aerosol population properties through instrument-specific measurements (Chen et al., 2013; Cheng et al., 2022; DeCarlo et al., 2006; Krueger et al., 2003; Wang, Pikridas, Spielman, et al., 2017; Zelenyuk et al., 2015). Other tools then target specific diagnostics such as optical properties, cloud activation, or ice nucleation (Alpert & Knopf, 2016; Petters & Kreidenweis, 2007; Sumlin et al., 2018). However, these components do not share a common internal representation of aerosol populations, which limits reuse of downstream analyses across sources. As a result, populations derived from different sources cannot be readily combined, compared, or analyzed using shared diagnostics.

`part2pop` complements rather than duplicates these tools. Its distinct contribution is a shared representation layer that standardizes heterogeneous aerosol populations and enables the same downstream diagnostics to be applied across them.

Software design

`part2pop` adopts a config-driven architecture centered on a small set of core data structures and registry-discovered builders (Figure 1). The primary structures are `AerosolSpecies`, `Particle`, and `ParticlePopulation`. `AerosolSpecies` defines intrinsic species properties such as density, molar mass, hygroscopicity, and surface-tension defaults, with optional overrides applied at build time. `ParticlePopulation` stores a species list, species-resolved particle masses, number concentrations, and particle identifiers. Individual particles are accessed and modified through the population interface. Together, these structures define a unified internal representation of aerosol composition, size, and abundance across heterogeneous sources.

The core representation is mass-based. This is a deliberate design trade-off: it is general enough to accommodate heterogeneous population sources while remaining simple enough to support consistent derived-property calculations. Population construction is handled by `PopulationBuilder` using configuration dictionaries (`population_cfg`), and diagnostic construction is handled by `VariableBuilder` using `var_cfg`. Both rely on registry-backed factory discovery, so new population builders and diagnostic modules can be added without modifying the core classes.

Current population types include model-derived populations from the Particle Monte Carlo model (`partmc`) and the four-mode Modal Aerosol Module (`mam4`) (Liu et al., 2016; Riemer et al., 2009), parameterized populations such as `monodisperse`, `binned_lognormals`, and `sampled_lognormals`, and observation-based populations such as `hiscale_observations` and `edx_observations`. The `hiscale_observations` builder reconstructs populations from complementary measurements collected during the HI-SCALE field campaign (Fast et al., 2019), including Fast Integrated Mobility Spectrometer (FIMS) size distributions, Aircraft-Integrated Meteorological Measurement System (AIMMS) aircraft-state data, miniSPLAT single-particle mass-spectrometer composition classes, and Aerosol Mass Spectrometer (AMS) bulk-composition constraints (Beswick et al., 2008; DeCarlo et al., 2006; Jayne et al., 2000; Wang, Pikridas, Spielman, et al., 2017; Wang, Pikridas, Pinterich, et al., 2017; Zelenyuk et al., 2015). The `edx_observations` builder reconstructs populations from particle size, class, and elemental-composition measurements collected on substrates using the Size and

Time-Resolved Aerosol Collector (STAC) and analyzed using scanning electron microscopy with energy-dispersive X-ray spectroscopy (SEM-EDX) or high-resolution focused ion beam/scanning electron microscopy (FIB/SEM) chemical imaging (Chen et al., 2013; Cheng et al., 2022; Krueger et al., 2003). STAC has been demonstrated for vertical aerosol sampling on the Atmospheric Radiation Measurement (ARM) user facility Tethered Balloon System (TBS) (Cheng et al., 2022).

Process-specific extensions such as `OpticalPopulation` and `FreezingPopulation` wrap the same base population representation rather than reimplementing population logic. This allows optical and freezing calculations to reuse the same underlying particle and population definitions while interfacing with established frameworks such as κ-Köhler theory and Mie scattering implementations (Alpert & Knopf, 2016; Petters & Kreidenweis, 2007; Sumlin et al., 2018). A visualization module and optional graphical viewer support exploratory analysis.

The architecture is also intentionally designed to be modifiable by both human developers and automated code-generation systems. Config-driven workflows, registry-based module discovery, minimal coupling between core data structures and analysis modules, and stable builder interfaces localize code changes. As a result, new population builders or diagnostics can be added without invasive edits to the core package, lowering the barrier to extension for both human developers and AI-assisted code generation.

DRAFT

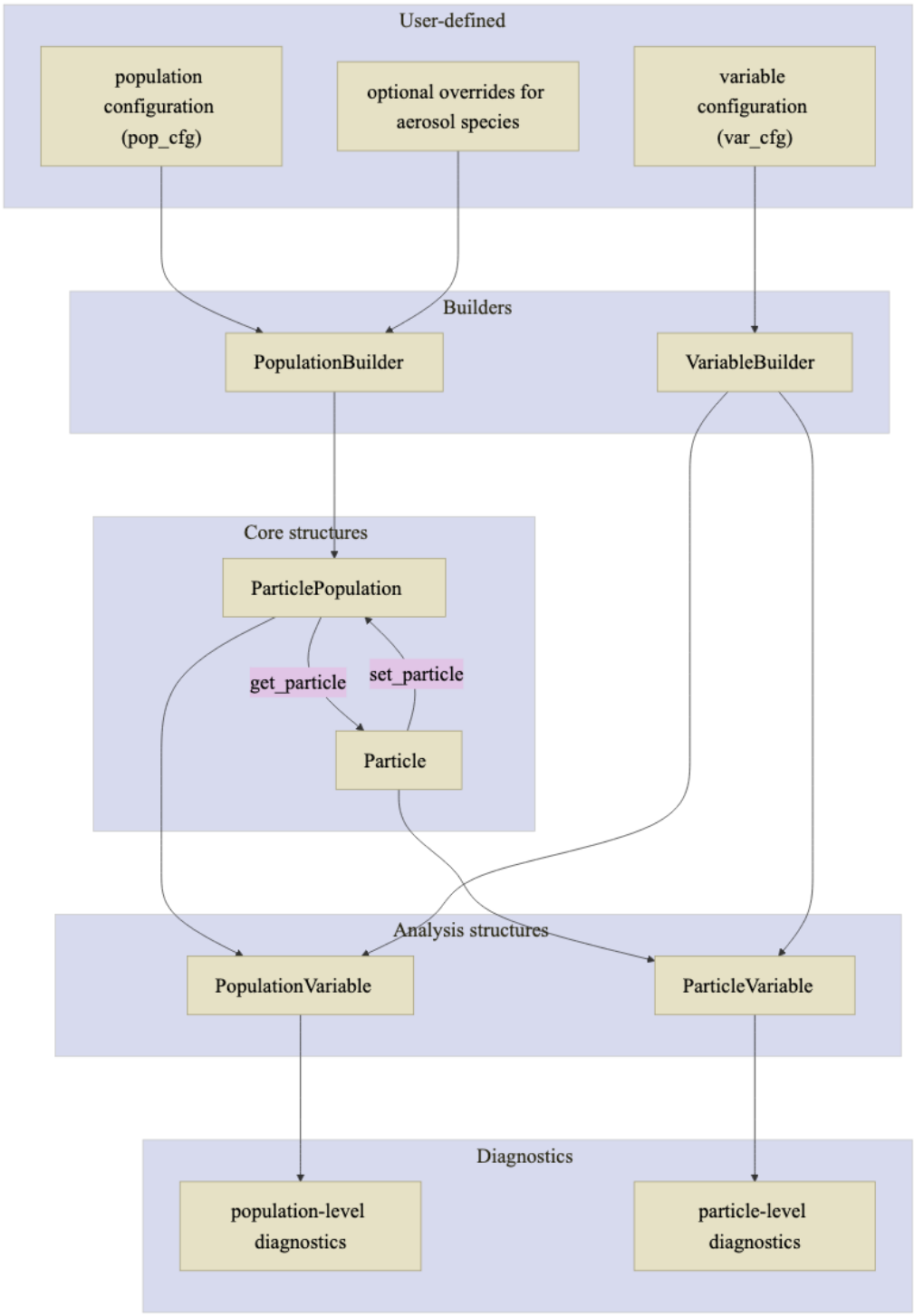

Figure 1: Overview of the config-driven architecture in `part2pop`. User-defined configuration files are processed by builder modules to construct standardized particle populations and derived analysis variables. Particle- and population-level diagnostics are then computed from these shared representations.

Research impact statement

part2pop targets an active methodological bottleneck in aerosol science: representation choice itself can introduce structural differences in aerosol diagnostics and can complicate comparison across models and datasets (Fierce et al., 2017, 2024; Fierce & McGraw, 2017). By making particle-resolved, modal, sectional, parameterized, and observation-based populations analyzable through a shared interface, the package provides reusable infrastructure for cross-representation comparison, diagnostic reuse, and initialization of downstream models.

The current release includes reproducible workflows for optical, CCN-relevant, freezing-related diagnostics for aerosol particle populations from a variety of sources. This positions part2pop as reusable research infrastructure rather than study-specific analysis code and supports its use in intercomparison and observation-constrained aerosol analysis.

AI usage disclosure

The core architecture, data model, and software design were developed by the authors without the use of generative AI tools. After the core structure was established, OpenAI's ChatGPT (versions 5.1–5.4) was used for code development and refactoring, development of the graphical user interface, and drafting and editing of documentation and manuscript text.

All AI-generated code and text were reviewed, modified, and verified by the authors, validated through testing and example workflows, and checked for scientific correctness. The software architecture was intentionally designed to support modification by automated code-generation tools, including large language models, through its modular and configuration-driven structure.

Acknowledgements

The part2pop package was developed under the Integrated Cloud, Land-surface, and Aerosol System Study (ICLASS), a Science Focus Area of the U.S. Department of Energy Atmospheric System Research program at Pacific Northwest National Laboratory (PNNL). Development of optics modules and links to the PartMC and MAM4 modules was supported by PNNL's Laboratory Directed Research and Development program. PNNL is a multi-program national laboratory operated for the U.S. Department of Energy by Battelle Memorial Institute under Contract No. DE-AC05-76RL01830.

Alpert, P. A., & Knopf, D. A. (2016). Analysis of isothermal and cooling-rate-dependent immersion freezing by a unifying stochastic ice nucleation model. *Atmospheric Chemistry and Physics*, *16*, 2083–2107. https://doi.org/10.5194/acp-16-2083-2016

Bauer, S. E., Wright, D. L., Koch, D., Lewis, E. R., McGraw, R., Chang, L.-S., Schwartz, S. E., & Ruedy, R. (2008). MATRIX (multiconfiguration aerosol TRacker of mIXing state): An aerosol microphysical module for global atmospheric models. *Atmospheric Chemistry and Physics*, *8*, 6003–6035. https://doi.org/10.5194/acp-8-6003-2008

Beswick, K. M., Gallagher, M. W., Webb, A. R., Norton, E. G., & Perry, F. (2008). Application of the aventech AIMMS20AQ airborne probe for turbulence measurements during the convective storm initiation project. *Atmospheric Chemistry and Physics*, *8*, 5449–5463. https://doi.org/10.5194/acp-8-5449-2008

Chen, H., Grassian, V. H., Saraf, L. V., & Laskin, A. (2013). Chemical imaging analysis of environmental particles using the focused ion beam/scanning electron microscopy technique: Microanalysis insights into atmospheric chemistry of fly ash. *Analyst*, *138*, 451–460. https://doi.org/10.1039/C2AN36318F

Cheng, Z., Liyu, A., Dexheimer, D., Lata, N. N., Kulkarni, G., Longbottom, C. M., Mei, F., & China, S. (2022). An automated size and time-resolved aerosol collector platform integrated

DRAFT

with environmental sensors to study the vertical profile of aerosols. *Environmental Science: Atmospheres*, *2*, 1263–1276. https://doi.org/10.1039/D2EA00097K

Cubison, M. J., Ervens, B., Feingold, G., Docherty, K. S., Ulbrich, I. M., Shields, L., Prather, K., Hering, S., & Jimenez, J. L. (2008). The influence of chemical composition and mixing state of Los Angeles urban aerosol on CCN number and cloud properties. *Atmospheric Chemistry and Physics*, *8*, 5649–5667. https://doi.org/10.5194/acp-8-5649-2008

DeCarlo, P. F., Kimmel, J. R., Trimborn, A., Northway, M. J., Jayne, J. T., Aiken, A. C., Gonin, M., Fuhrer, K., Horvath, T., Docherty, K. S., Worsnop, D. R., & Jimenez, J. L. (2006). Field-deployable, high-resolution, time-of-flight aerosol mass spectrometer. *Analytical Chemistry*, *78*(24), 8281–8289. https://doi.org/10.1021/ac061249n

Fast, J. D., Berg, L. K., Alexander, L., Bell, D., D'Ambro, E., Hubbe, J., Kuang, C., Liu, J., Long, C., Matthews, A., Mei, F., Newsom, R., Pekour, M., Pinterich, T., Schmid, B., Schobesberger, S., Shilling, J., Smith, J. N., Springston, S., … Zelenyuk, A. (2019). Overview of the HI-SCALE field campaign: A new perspective on shallow convective clouds. *Bulletin of the American Meteorological Society*, *100*(5), 821–840. https://doi.org/10.1175/BAMS-D-18-0030.1

Fierce, L., Bond, T. C., Bauer, S. E., Mena, F., & Riemer, N. (2016). Black carbon absorption at the global scale is affected by particle-scale diversity in composition. *Nature Communications*, *7*, 12361. https://doi.org/10.1038/ncomms12361

Fierce, L., & McGraw, R. L. (2017). Multivariate quadrature for representing cloud condensation nuclei activity of aerosol populations. *Journal of Geophysical Research: Atmospheres*, *122*(18), 9867–9878. https://doi.org/10.1002/2016JD026335

Fierce, L., Onasch, T. B., Cappa, C. D., Mazzoleni, C., China, S., Bhandari, J., Davidovits, P., Fischer, D. A., Helgestad, T. J., Lambe, A. T., Sedlacek, I., Arthur J., Smith, G. D., & Wolff, L. (2020). Radiative absorption enhancements by black carbon controlled by particle-to-particle heterogeneity in composition. *Proceedings of the National Academy of Sciences of the United States of America*, *117*(10), 5196–5203. https://doi.org/10.1073/pnas.1919723117

Fierce, L., Riemer, N., & Bond, T. C. (2013). When is cloud condensation nuclei activity sensitive to particle characteristics at emission? *Journal of Geophysical Research: Atmospheres*, *118*(24), 13476–13488. https://doi.org/10.1002/2013JD020608

Fierce, L., Riemer, N., & Bond, T. C. (2017). Toward reduced representation of mixing state for simulating aerosol effects on climate. *Bulletin of the American Meteorological Society*, *98*(5), 971–980. https://doi.org/10.1175/BAMS-D-16-0028.1

Fierce, L., Yao, Y., Easter, R. C., Ma, P.-L., Sun, J., Wan, H., & Zhang, K. (2024). Quantifying structural errors in cloud condensation nuclei activity from reduced representation of aerosol size distributions. *Journal of Aerosol Science*, *181*, 106388. https://doi.org/10.1016/j.jaerosci.2024.106388

Jayne, J. T., Leard, D. C., Zhang, X., Davidovits, P., Smith, K. A., Kolb, C. E., & Worsnop, D. R. (2000). Development of an aerosol mass spectrometer for size and composition analysis of submicron particles. *Aerosol Science and Technology*, *33*(1–2), 49–70. https://doi.org/10.1080/027868200410840

Krueger, B. J., Grassian, V. H., Iedema, M. J., Cowin, J. P., & Laskin, A. (2003). Probing heterogeneous chemistry of individual atmospheric particles using scanning electron microscopy and energy-dispersive x-ray analysis. *Analytical Chemistry*, *75*(19), 5170–5179. https://doi.org/10.1021/ac034455t

Lata, N. N., Zhang, B., Schum, S., Mazzoleni, L. R., Brimberry, R. C., Marcus, M. A., Cantrell, W. H., Fialho, P., Mazzoleni, C., & China, S. (2021). Aerosol composition, mixing state, and phase state of free tropospheric particles and their role in ice cloud formation. *ACS Earth and*

DRAFT

Space Chemistry, *5*(12), 3499–3510. https://doi.org/10.1021/acsearthspacechem.1c00315

Liu, X., Ma, P.-L., Wang, H., Tilmes, S., Singh, B., Easter, R. C., Ghan, S. J., & Rasch, P. J. (2016). Description and evaluation of a new four-mode version of the modal aerosol module (MAM4) within version 5.3 of the community atmosphere model. *Geoscientific Model Development*, *9*, 505–522. https://doi.org/10.5194/gmd-9-505-2016

Mann, G. W., Carslaw, K. S., Ridley, D. A., Spracklen, D. V., Pringle, K. J., Merikanto, J., Korhonen, H., Schwarz, J. P., Lee, L. A., Manktelow, P. T., Woodhouse, M. T., Schmidt, A., Breider, T. J., Emmerson, K. M., Reddington, C. L., Chipperfield, M. P., & Pickering, S. J. (2012). Intercomparison of modal and sectional aerosol microphysics representations within the same 3-D global chemical transport model. *Atmospheric Chemistry and Physics*, *12*(10), 4449–4476. https://doi.org/10.5194/acp-12-4449-2012

Petters, M. D., & Kreidenweis, S. M. (2007). A single parameter representation of hygroscopic growth and cloud condensation nucleus activity. *Atmospheric Chemistry and Physics*, *7*(8), 1961–1971. https://doi.org/10.5194/acp-7-1961-2007

Riemer, N., Ault, A. P., West, M., Craig, R. L., & Curtis, J. H. (2019). Aerosol mixing state: Measurements, modeling, and impacts. *Reviews of Geophysics*, *57*(2), 187–249. https://doi.org/10.1029/2018RG000615

Riemer, N., West, M., Zaveri, R. A., & Easter, R. C. (2009). Simulating the evolution of soot mixing state with a particle-resolved aerosol model. *Journal of Geophysical Research: Atmospheres*, *114*, D09202. https://doi.org/10.1029/2008JD011073

Stevens, R., & Dastoor, A. (2019). A review of the representation of aerosol mixing state in atmospheric models. *Atmosphere*, *10*(4), 168. https://doi.org/10.3390/atmos10040168

Sumlin, B. J., Heinson, W. R., & Chakrabarty, R. K. (2018). Retrieving the aerosol complex refractive index using PyMieScatt: A Mie computational package with visualization capabilities. *Journal of Quantitative Spectroscopy and Radiative Transfer*, *205*, 127–134. https://doi.org/10.1016/j.jqsrt.2017.10.012

Tilmes, S., Mills, M. J., Zhu, Y., Bardeen, C. G., Vitt, F., Yu, P., Fillmore, D., Liu, X., Toon, O. B., & Deshler, T. (2023). Description and performance of a sectional aerosol microphysical model in the community earth system model (CESM2). *Geoscientific Model Development*, *16*, 6087–6125. https://doi.org/10.5194/gmd-16-6087-2023

Wang, J., Pikridas, M., Pinterich, T., Spielman, S. R., Tsang, T., McMahon, A., & Smith, S. (2017). A fast integrated mobility spectrometer for rapid measurement of sub-micrometer aerosol size distribution, part II: Experimental characterization. *Journal of Aerosol Science*, *113*, 119–129. https://doi.org/10.1016/j.jaerosci.2017.05.001

Wang, J., Pikridas, M., Spielman, S. R., & Pinterich, T. (2017). A fast integrated mobility spectrometer for rapid measurement of sub-micrometer aerosol size distribution, part I: Design and model evaluation. *Journal of Aerosol Science*, *108*, 44–55. https://doi.org/10.1016/j.jaerosci.2017.02.012

Zaveri, R. A., Easter, R. C., Fast, J. D., & Peters, L. K. (2008). Model for simulating aerosol interactions and chemistry (MOSAIC). *Journal of Geophysical Research: Atmospheres*, *113*, D13204. https://doi.org/10.1029/2007JD008782

Zelenyuk, A., Imre, D., Wilson, J. M., Zhang, Z., Wang, J., & Mueller, K. (2015). Airborne single particle mass spectrometers (SPLAT II and miniSPLAT) and new software for data visualization and analysis in a Geo-Spatial context. *Journal of the American Society for Mass Spectrometry*, *26*(2), 257–270. https://doi.org/10.1007/s13361-014-1043-4

DRAFT